Ising Model

Joey Bryson
Level 3 MSci. Laboratory, School of Physics, University of Bristol
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The critical behavior of the two-dimensional Ising Model was investigated through Monte Carlo simulations, using the Swendsen-Wang algorithm. Finite-size scaling analysis and Binder Cumulant intersections were used to estimate critical exponents and the critical temperature. The results yielded values for the exponents $\nu = 1.000 \pm 0.064$ and $\gamma = 1.76 \pm 0.12$, as well as a Binder estimate of critical temperature $T_c = 2.26900 \pm 0.00027$, in close agreement with Onsager's exact solutions[1]. Despite some systematic deviations for smaller system sizes, the findings validate the theoretical power-law relationships and demonstrate the effectiveness of finite-size scaling techniques in measuring phase transition phenomena.

I. INTRODUCTION

The two-dimensional Ising model in one of the first and simplest models found exhibits a continuous phase transition. First introduced over in 1925 by Ernst Ising[2] and then shown to have a phase transition in 1936 by Rodolf Peierls[3]. The study of continuous phase transition phenomena is broad and far-reaching, with applications extending as far as its initial inception as a model of ferromagnetism to biological processes like the transition between wakefulness and sleep in animals[4]. Study of the Ising Model can provide great insight due to its analytic simplicity relative to other models and phenomena which it's behavior can describe.

II. THEORY

Start be defining a two-dimensional square lattice of side length N where each lattice site has a corresponding 'spin' $\sigma_i \in \{-1, +1\}$. The Ising Model explored in this experiment can be defined simply by the Hamiltonian of the spin micro-state $\boldsymbol{\sigma}$:

$$H(\boldsymbol{\sigma}) = -\sum_{\langle ij \rangle} \sigma_i \sigma_j \tag{1}$$

where $\langle ij \rangle$ indicates that the the spin pairs considered are adjacent. We allow for edge sites to be adjacent to satisfy periodic boundary conditions. In order to produce distributions for the above Hamiltonian in the thermodynamic limit, any computational algorithm must satisfy the condition of detailed balance.

$$\frac{P(\boldsymbol{\mu} \to \boldsymbol{\nu})}{P(\boldsymbol{\nu} \to \boldsymbol{\mu})} = e^{-\beta(H(\boldsymbol{\nu}) - H(\boldsymbol{\mu}))} \tag{2}$$

That is that the ratio of transitioning to and from any two micro-states $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$ is proportional to the ratio of their Boltzmann factors. In this experiment quantities are given in their natural units and so $\beta = 1/T$, this has no effect on the fundamentally unit-less Critical Exponents explored later. In the study of the critical phenomena of the Ising Model a key variable which can be used to understand the system's behavior is the correlation length ξ, $\langle (\sigma_i - \langle \sigma_i \rangle)(\sigma_j - \langle \sigma_j \rangle) \rangle \sim e^{-\frac{|x-y|}{\xi}}$ where x and y are the spacial position vectors of the spins σ_i and σ_j. The correlation length characterizes the typical maximum size of clusters formed, as it defines a typical length scale within which, spins are likely to be aligned. In an infinite lattice it is known that the correlation length increases asymptotically as the system approaches criticality. This is reflected in the power law around criticality

$$\xi \sim |T - T_c|^{-\nu} \tag{3}$$

If the correlation length goes to infinity at the critical temperature, this implies that there is no characteristic size for structure in the system. In other words, the system is scale invariant and has fractal symmetries. Even at an infinite scale, the system does not look homogeneous. This leads to similar divergent behavior of the other quantities in the system. Additional to the one already seen for correlation length there three other standard power law relationships around the critical temperature

$$M \sim |T - T_c|^{\beta} \quad \chi \sim |T - T_c|^{-\gamma} \quad C \sim |T - T_c|^{-\alpha} \tag{4}$$

Where $M = \sum_{i=1}^{N^2} \langle \sigma_i \rangle$ is magnetization or more generally for non magnetic contexts the 'order parameter', $\chi = \frac{1}{k_B T} \left(\langle M^2 \rangle - \langle M \rangle^2 \right)$ is the susceptibility and $C = \frac{1}{k_B T^2} \left(\langle H^2 \rangle - \langle H \rangle^2 \right)$ is the heat capacity. Together, ν, β, γ and α are known as the critical exponents and were analytically solved by Onsager[1] to be $1, \frac{1}{8}, \frac{7}{4}$ and 0 respectively. It turns out these exponents arise from the dimensionality and symmetries of the system and they can arise systems with different microscopic behavior, this allows phase transitions in different contexts to be grouped into Universality Classes by methods such as renormalization group flow[5]. The study of critical exponents through models like the Ising Model can have incredible descriptive power in real (supposedly) unrelated physical systems. For example the three dimensional counterparts to the exponents above have been found in fluid's critical behavior. [6]

In a finite size Ising Model system, such as one that is possible to simulate, ξ is limited proportional to the system size, because points start becoming correlated with

themselves across the periodic boundary so $\xi_{max} \propto N$. This means that a true critical point is not reached, instead, there is a Pseudo-critical temperature as a function of system size N, $T_c(N)$ at which ξ_{max} is reached. Up to this breaking point, the power laws described earlier still govern the behavior of the key parameters provided N is large enough that $T_c(N)$ is sufficiently close to $T_c(\infty) = T_c$. So we have

$$N \propto \xi_{max} \sim |T_c(N) - T_c|^{-\nu} \tag{5}$$

Since the correlation length is the mechanism for the behavior of the other parameters in the critical region, it can be observed that the following holds.

$$\xi \propto M^{-\frac{\beta}{\nu}} \propto \chi^{\frac{\gamma}{\nu}} \tag{6}$$

in the finite regime. It can be seen that N and the values of the other key quantities in the critical region obey power laws, and given data of both, they can be plotted and fitted on a log-log scale to derive estimates for the critical exponents. These techniques are known as finite size scaling.

Another key quantity solved by Onsager for the Ising Model without an external field is the critical temperature. It was found to be $T_c = \frac{1}{\ln(1+\sqrt{2})}$. A method to estimate this through simulation involves the use of a parameter known as the Binder Cumulant introduced by Kurt Binder in 1981[7] defined by

$$U_N = 1 - \frac{\langle \sigma^4 \rangle}{3\langle \sigma^2 \rangle^2} \tag{7}$$

At the critical temperature, the relationship above shows magnetization per spin follows $\langle |\sigma| \rangle \propto N^{-\beta/\nu}$. This implies a finite-size scaling form for the distribution function, $P_N(\sigma) = N^{\beta/\nu} P(u)$, where $u = \sigma N^{\beta/\nu}$. The Binder cumulant can be evaluated using the transformation $\sigma = uN^{-\beta/\nu}$, leading to $\langle \sigma^2 \rangle^2 \propto N^{-4\beta/\nu}$ and $\langle \sigma^4 \rangle \propto N^{-4\beta/\nu}$. Consequently, U_N becomes independent of N at the critical point, making it a precise tool for estimating T_c by identifying the temperature where curves of U_N for different N intersect, provided system sizes are large enough to mitigate for microscopic effects.

III. ALGORITHMS

. Two algorithms for simulating the Ising Model were explored in the course of this experiment - the Metropolis-Hastings(MH) algorithm, and the Swendsen-Wang algorithm. The MH algorithm was first investigated due to the simplicity of its implementation. A single spin flip is considered at a time and the detailed balance is simple to demonstrate[8]. A useful parameter when considering an algorithm for a thermodynamic model is the relaxation time τ, defined similarly to ξ but in the dimension of simulation time steps. It can be thought of as the number of steps it takes to forget a previous state, in order to produce equilibrium statistics the number of simulation steps $n \gg \tau$. τ is related to ξ by $\tau \propto \epsilon^z$ with z known as the Dynamical Critical Exponent(DCE) so at Pseudo-critical point $\tau \propto N^z$. It is known that DCE for the MH algorithm is approximately 2.1665[9], this leads to the requirement of incredibly long simulation time, especially for large simulations which are required to minimize microscopic effects. An alternative algorithm more suited to the simulation of critical behavior is the Swendsen-Wang(SW) algorithm[10] which has been found to have a DCE in the neighborhood of 0.2 to 0.35[11].This is the result of the mechanism of the Swendsen-Wang algorithm, it acts on bond clusters formed on the order of ξ. The mechanism and detailed balance proof of the SW algorithm is not within the scope of this report however this can be found in the experimental lab book.

For determining the critical exponents and temperature the SW algorithm was used. First a random Bernoulli-1/2 spin state was chosen, then a set of unrecorded 'equilibrium steps'(ES) were carried out, then a much larger set of 'measurement steps'(MS) were carried out and at each step the magnetization and energy of the state were recorded and saved, the distribution of these could be used to determine M, χ and C. This was performed over a range of system sizes and temperatures (around the critical region) depending on the ranges required for the analyses.

IV. RESULTS

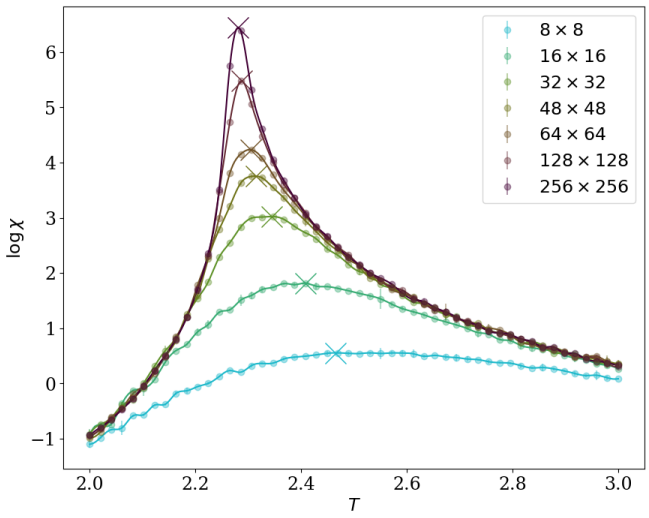


FIG. 1. Log Susceptibility against temperature for different system sizes. Spline fits are used to interpolate the data and calculated peaks are indicated by X symbols. It can be thought of that there is an ideal $N \to \infty$ asymptotic relation which systems approach as N increases. The low N peaks are much flatter and so have much greater uncertainty in $T_c(N)$

Magnetization data was collected for systems of sizes $8 \leq N \leq 256$, for 50 temperatures $2.0 \leq T \leq 3.0$, with 1000 ES and 10000 MS. χ was calculated for each T and N. Spline fits were used to approximate the functions for $\log(\chi)$ against temperature T. Then a Savitzky–Golay filter was used to smooth the data and approximate the peaks χ_{max}. Errors in the peaks were acquired via a bootstrap method. By (5) and (6) the temperature at which these occur can then be taken as the Pseudo-Critical temperature $T_c(N)$ for each system size. $T_c(N)$ against N was then fitted with a function of the form $T_c^*(N^*) = AN^{*-1/B} + C$ where A, B, and C were fitting parameters and $-1/B$ and C could be used as estimates for ν and Tc respectively. Additionally, an estimate for ratio $\frac{\gamma}{\nu}$ and therefore γ could be found from the relation $\log \chi_{M,\text{peak}} = \frac{\gamma}{\nu} \log N$. The estimated values are given in Table I.

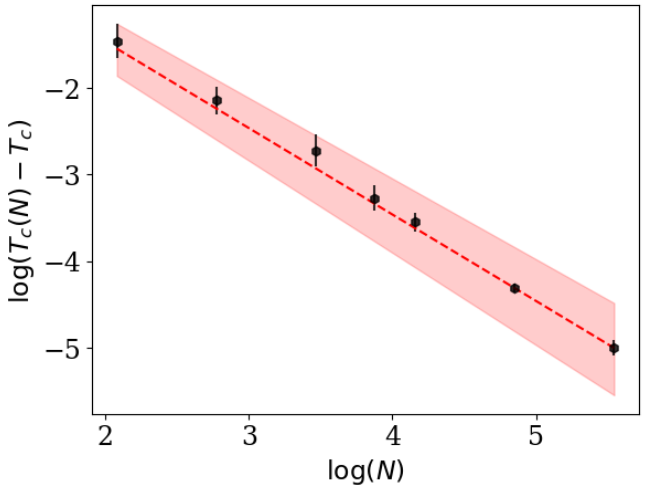


FIG. 2. Plot of PC temperature data points $log(T_c(N) - C)$, and relation $log(T_c(N) - C) = log(A) - Blog(N))$ where A, B and C are the fitted parameters. The red shaded region indicates the uncertainty in the fit parameters A and B

Another set of data was collected with the same size range with 25 temperatures in a tighter range of $2.55 \leq T \leq 2.85$ with 1000 ES and 50000 MS. The Binder Cumulant U_N was calculated for each temperature and system size, the data-points of U_N vs T were then fitted using smoothing spline fits. The intersection temperatures T_i of consecutive pairs of fits(or the zeros of their sums $D_i = U_{N_i} + U_{N_{i+1}}$) were then found and their errors given by $\Delta D / |\frac{dD}{dT}|_{T_{\text{intersection}}}$, with $\Delta U_{N,i}$ given from the spline fitting algorithm. Finally, an estimate for T_c was given by a weighted sum, with the weights being the inverse of the variance. This gave an estimate of $T_c = 2.26900 \pm 0.00027$

V. DISCUSSION

The results from the fist data set had an interesting asymmetry, computed ν had a very accurate value but a likely overestimated uncertainty and T_c had a very much more inaccurate value but a underestimated uncertainty uncertainty. This indicates that the the peak data was non-theoretic to the fit $T_c^*(N^*) = AN^{*-1/B} + C$. This could be caused by a systematic shift in the peak position for smaller system sizes, either as a result of microscopic effects or the method of peak finding. The three parame-

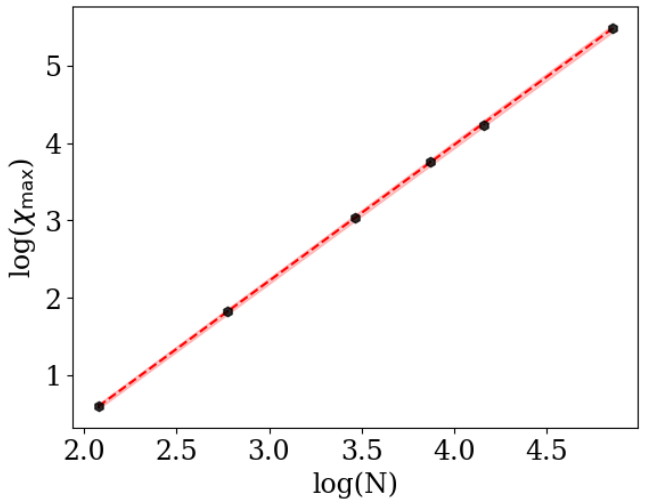


FIG. 3. Plot of $log(\chi_{max}$ against $log(N)$ and the fit $log(\chi_{max} = Alog(N)$the relationship showing low relative variance compared to the previous graph

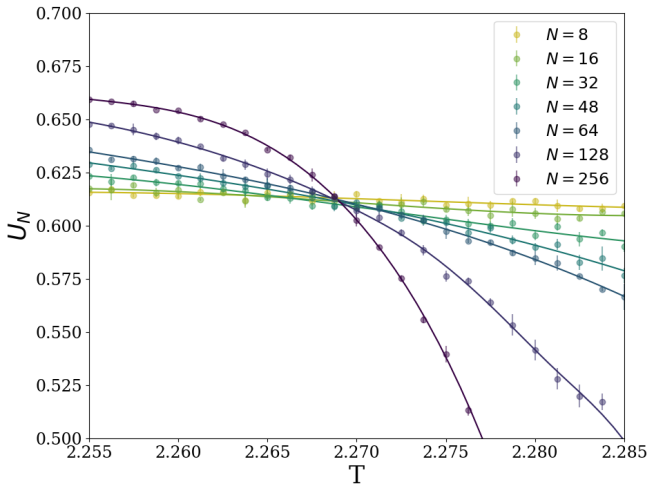


FIG. 4. U_N against temperature T for different system sizes, with corresponding spline fits

Critical variable	Experimental value
ν	1.000 ± 0.064
γ	1.76 ± 0.12
T_c	2.2742 ± 0.0006

TABLE I. Experimentally determined critical exponents.

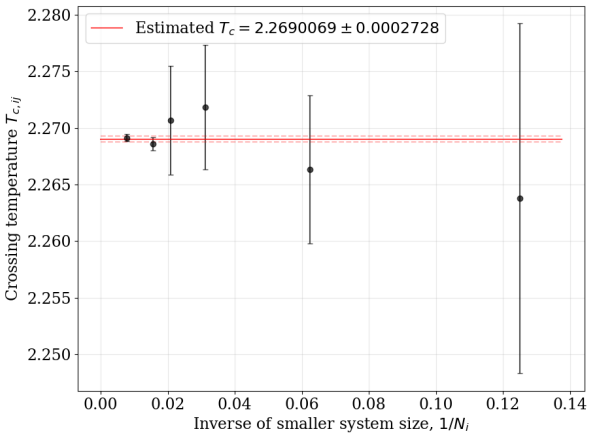


FIG. 5. Intersection temperature of successive fits of U_{N_i} and $U_{N_{i+1}}$ against $1/N_i$, with weighted mean T_c. The graph shows convergence of $T_{c,i}$ as N increases since errors decrease, however, no clear relationship of the $T_{c,i}$ vs $1/N$ that could be extrapolated to $1/N = 0$ is obvious.

ter model, considering there are only six data points, also introduces the possibility of overfitting, although this is deemed unlikely given the repeatability of results. The known value T_c was also substituted into the fitting function, however, this yielded a worse estimate for ν and a less linear (upwards curving) relationship in the plot shown in Fig. 2, this is further evidence there was a theoretic or methodical error. Interestingly, the relationship between the peak height and system size fit theoretical expectations very well. This lent to a small error on the ratio γ/μ and the shown error for γ is predominantly a propagation from the error in μ.

The method used to find T_c from the second dataset produced a good value with a reasonable error. The majority of the contribution to the final value came from the largest system crossing points as their errors were much smaller. The fact each error bar overlaps the final estimate is a good indication of the validity of the method to determine the error in $T_{c,i}$ as each value individually should be a valid estimate for T_c, assuming finite size effects on U_N have minuscule contribution (which is indicated by Fig. 5).

One possible improvement could be to use more specific forms of fitting for the Susceptibility and Binder Cumulant. Analytic functions could be derived which approximate the behavior of these values for finite sizes. Then parameters for these functional forms could be found by general curve fitting algorithms. This could provide much more precise values, especially for the peak positions of χ, since spline fitting assumes local low order polynomial behavior, and there are too few data points over each peak for this to be corrected for. Additionally this would address overfitting by spline functions by encoding a the specific complexity of the function required.

VI. CONCLUSION

In this experiment, the critical behavior of the two-dimensional Ising Model was analyzed using Monte Carlo simulations with the Swendsen-Wang algorithm. Our results successfully showed the expected power-law relationships for key thermodynamic quantities near the critical point. The estimated values for the critical exponents μ and γ were in agreement with theoretical predictions, though some systematic biases were observed, particularly for the first estimation of . The second estimation of via the Binder Cumulant provided an accurate value with a reasonable error. Overall, the findings affirm the theoretical expectations and support the utility of finite-size scaling and the Binder Cumulant in computational determination of critical values.

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